

Placement Brochure 2026-27



Photonics Science and Engineering Program (PSE)

Centre for Lasers and Photonics(CELP),
Indian Institute of Technology Kanpur

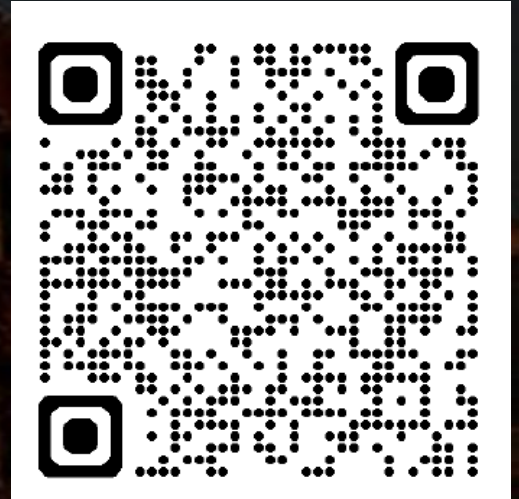
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About Us

The Center for Lasers and Photonics (CELP) at IIT Kanpur is an interdisciplinary center of excellence that integrates cutting-edge advances in engineering, physics, and computational science to push the frontiers of Photonics Science and Engineering (PSE). CELP bridges the gap between theory and application, with a focus on developing innovative solutions that span a broad range of domains — including optoelectronics, integrated photonic systems, high-speed communication, quantum technologies, and AI-assisted photonic design.

CELP offers three postgraduate programs: M.Tech, M.S. (Research), and Ph.D. in Photonics Science and Engineering. These programs are uniquely structured to equip students with both theoretical depth and hands-on expertise in areas such as semiconductor optoelectronics, fiber optics, optical communication systems, photonic integrated circuits, signal processing, and machine learning applications in photonics and imaging.

Graduates from CELP are well-positioned in diverse sectors, including telecom, data networking, VLSI design, embedded systems, AI-driven imaging, fiber and quantum optics. They have contributed significantly as entrepreneurs, researchers, and technology leaders in organizations like ISRO, DRDO, C-DOT, Intel, Nvidia, Sterlite Technologies, and Qualcomm. With a strong foundation in electronics, communication systems, and computational modeling, the programs at CELP aim to develop the next generation of engineers and scientists capable of shaping the future of photonic and allied technologies.



Watch our Department's video

Welcome Message from the Head



Prof. Pradeep Kumar Krishnamurthy,
Head, PSE(CELP), IIT Kanpur

For the placement season 2025–26, we warmly invite industry leaders to recruit students from our department.

This is an excellent opportunity to engage with a highly skilled and multidisciplinary talent pool trained at the intersection of cutting-edge science and advanced engineering applications. The program is uniquely designed to equip students with a strong foundation in photonics, electronics, electrical engineering, and applied physics, while also emphasizing integration with software and hardware systems relevant to today's technology-driven industries.

Through a balanced curriculum and rigorous hands-on training, our students are prepared to address complex real-world challenges across sectors like telecommunications, VLSI, embedded systems, high-speed optics, and AI-enabled imaging. Our coursework is aligned with industry standards, and students gain practical exposure through industry-oriented projects carried out in our state-of-the-art laboratories. They develop not only strong technical skills but also critical thinking, adaptability, and a problem-solving mindset.

This year we are also opening our project portfolio directly to industry: several of the projects in the following pages map onto problems our recruiters' product teams are actively working on, and we welcome collaboration, sponsored projects and internships alongside placement.

Curriculum

Core Courses

- Introduction to Photonics(PSE601A)
- Principles of Lasers and Detectors(PSE602A): Generation, Amplification and Detection of light
- Numerical Methods in Optics(PSE603A)
- Photonic Systems and Applications(PSE604A): Lasing Systems, Mode Locking, Fiber Optics, Non-linear Optics
- Photonics Lab Techniques(PSE605A)
- Research in Photonics and Lasers(PSE606A)
- Special Topics in Photonics(PSE607A)

Department Electives

- Nanophotonics(EE798I)
- Semiconductor Optical Communication Devices(EE685)
- Fiber Optic Communication(EE684)
- Quantum and Wave Phenomena(EE683)
- Signal Processing for Optical Metrology(EE798X)
- Advanced Fiber-optic Communications(EE798U)
- Optical Coherent Imaging(EE620)
- Kalman Filtering And Its Applications(EE657)
- Quantum Computing(CHM696)

Open Electives

- VLSI System Design(EE619)
- Mixed Signal IC Design(EE698I)
- Deep Learning For Computer Vision(CS776)
- Quantum Computing And Communication(EE798V)
- RFIC Design(EE698L)
- Analog/Digital VLSI Circuits(EE610)
- IC Fabrication Technology(EE618)
- Computer Vision and Deep Learning(EE655)
- 5G Wireless Technologies(EE698Q)
- Advanced Topics in Machine Learning(EE698R)
- High Frequency Analog Circuit Design(EE613)
- Parallel Computing(CS633)
- Atomic, Molecular & Optical Physics(PHY612)



On-going Projects

Sponsored and departmental research currently running in CELP laboratories.

Frequency-Coded Quantum Key Distribution over 25-km Fibre-Optic Links

Development of frequency-coded QKD solutions suitable for deployment on 25-km optical fibre links.

Electro-Optic & Magneto-Optic Devices for High-Speed Optical Distribution

Electro-optic and magneto-optic interaction-based high-speed signal distribution.

Photodiode Arrays for Near-Infrared Detection and Tracking

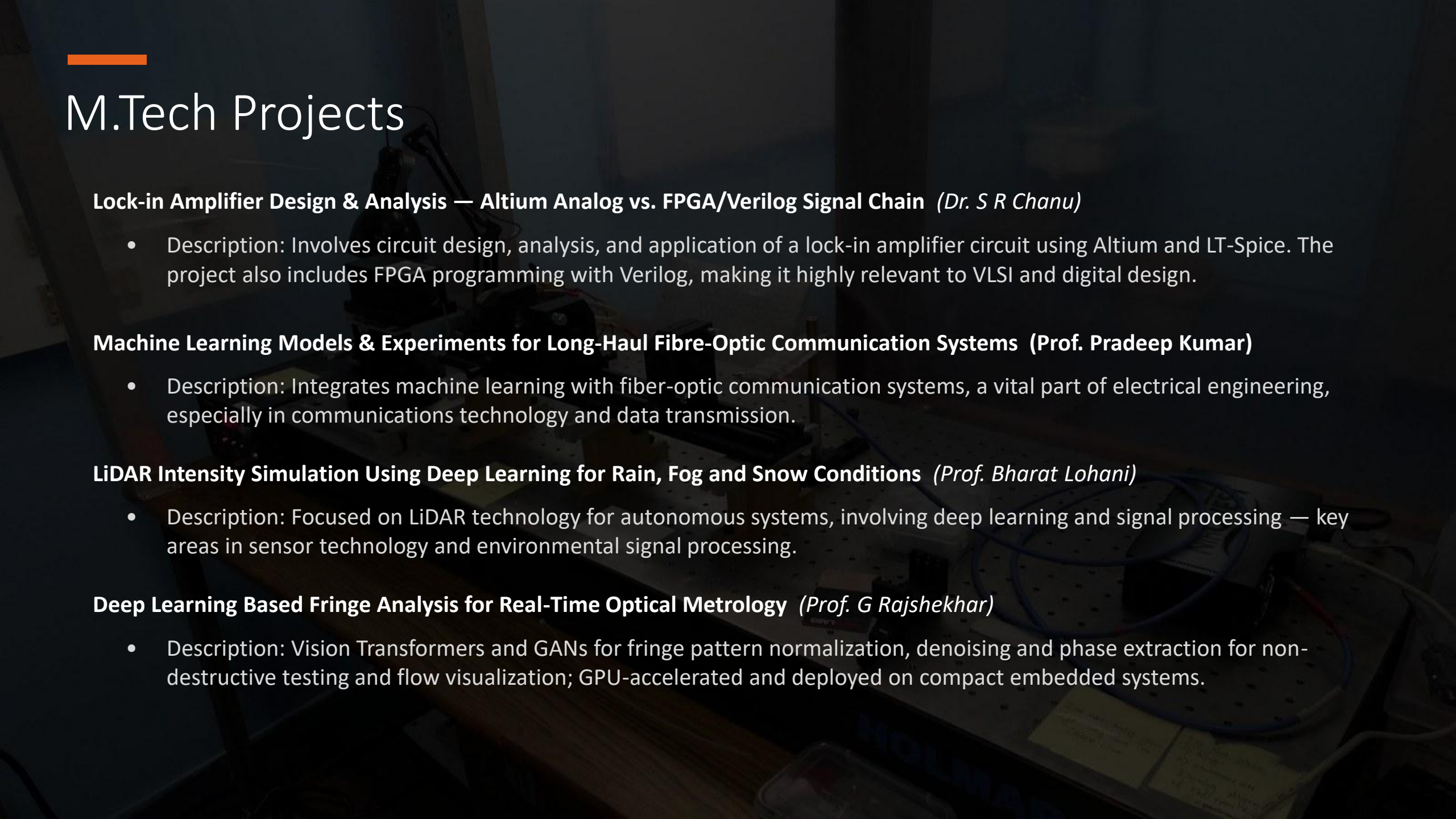
NIR-sensitive photodiode array design and characterisation for detection and tracking.

Nanoparticle-Based Photonic Elements — Design, Fabrication and Characterization

Full design–fabrication–characterisation cycle for nanoparticle-based photonic elements.

RTE-Tomography-Based Cloud Monitoring for Atmospheric Retrieval

Radiative-transfer-equation tomography applied to satellite cloud monitoring and retrieval.



M.Tech Projects

Lock-in Amplifier Design & Analysis — Altium Analog vs. FPGA/Verilog Signal Chain *(Dr. S R Chanu)*

- Description: Involves circuit design, analysis, and application of a lock-in amplifier circuit using Altium and LT-Spice. The project also includes FPGA programming with Verilog, making it highly relevant to VLSI and digital design.

Machine Learning Models & Experiments for Long-Haul Fibre-Optic Communication Systems *(Prof. Pradeep Kumar)*

- Description: Integrates machine learning with fiber-optic communication systems, a vital part of electrical engineering, especially in communications technology and data transmission.

LiDAR Intensity Simulation Using Deep Learning for Rain, Fog and Snow Conditions *(Prof. Bharat Lohani)*

- Description: Focused on LiDAR technology for autonomous systems, involving deep learning and signal processing — key areas in sensor technology and environmental signal processing.

Deep Learning Based Fringe Analysis for Real-Time Optical Metrology *(Prof. G Rajshekhar)*

- Description: Vision Transformers and GANs for fringe pattern normalization, denoising and phase extraction for non-destructive testing and flow visualization; GPU-accelerated and deployed on compact embedded systems.



M.Tech Projects

FPGA Implementation of Fast Hash Functions & Oracles for Quantum Random Number Generators *(Prof. Pradeep Kumar)*

- Description: Focuses on FPGA implementation, directly related to digital hardware design — an essential area of VLSI. Leverages FPGA for functions beneficial to cryptographic applications.

Photonic Devices for Quantum Technology Applications *(Prof. Shilpi Gupta)*

- Description: Fabrication and characterisation of integrated photonic components for quantum information processing and quantum communication.

Electromagnetic Stealth for Defence Applications *(Prof. R Vijaya)*

- Description: Micro-nano photonics including simulation, fabrication and testing — relevant to electromagnetic design and signal processing.

Laser Flash Photolysis Study of Carrier Dynamics in Perovskites *(Prof. Pratik Sen)*

- Description: Centered on carrier dynamics in perovskite materials, relevant for optoelectronics and semiconductor applications.

Constructing a Nano-viscometer with Optical Tweezers *(Prof. Debabrata Goswami)*

- Description: Optical tweezers with enhanced sensitivity using femtosecond laser tweezers for single-particle micro-rheology.

Lab Facilities

Opto-electronics and Nano-Fabrication Laboratory

- Multiquantum well Intermixed Waveguide Grating Assisted Couplers
- High Speed Waveguide Photodiodes
- Photodiode Arrays
- SiGeC/SiC Quantum Dash LEDs on silicon by spin-on technique
- Laser assisted vision through Fog

Cold ion Quantum Technologies Lab

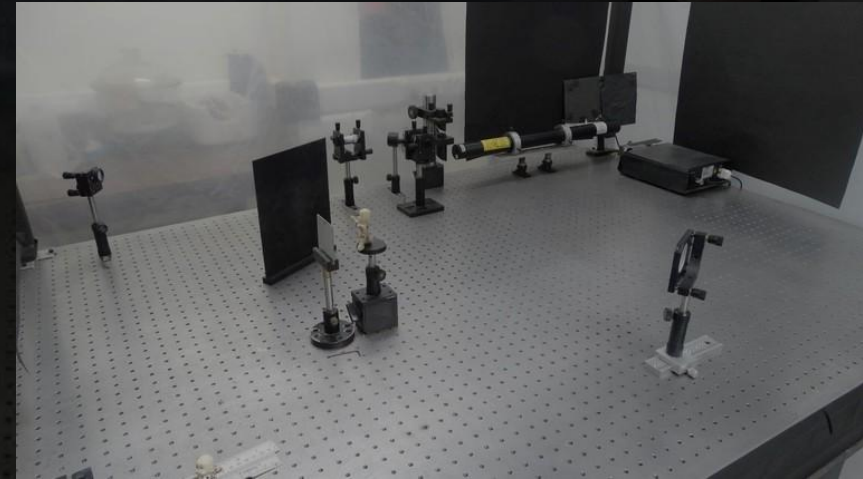
- Quantum simulation
- Cold ion trapping
- FPGA based on Laser experiment

Optical Metrology and Imaging Laboratory

- Digital holography
- Machine learning algorithms for optical metrology
- High performance techniques

Photonics Devices Lab

- Flat devices
- Quantum Communications
- Non-Invasive Optical technology Flexible Photonics
- Opto-electronic devices



Optical Metrology and Imaging Lab



Opto-electronics and nano-fabrication Lab

Lab Facilities

Femto Lab

- Quantum Computing
- Pulse Shaping
- Optical Tweezers
- Thermal Lensing and Imaging

Laser Fabrication Lab

- High speed Flow Visualization
- Emission spectroscopy

Photonics Lab

- Photonics
- Nonlinear Optics Fiber Optics Photonic crystals
- Optical Nanostructures

Diffuse Image Laboratory

- Quantum Optics
- Non linear Optics
- Photonics Bandgap materials
- Metamaterials



Laser fabrication Lab



Femto Lab

Lab Facilities

Optoelectronics & Nano-Fabrication Lab

- Multi quantum Well Devices: Developing intermixed waveguide grating assisted couplers.
- High-Speed Photodiodes: Fabricating photodiode arrays for high-bandwidth comms.
- Silicon Photonics: Demonstrating SiGeC/Si quantum dash LEDs via spin-on techniques.

Femtosecond Laser Spectroscopy Laboratory

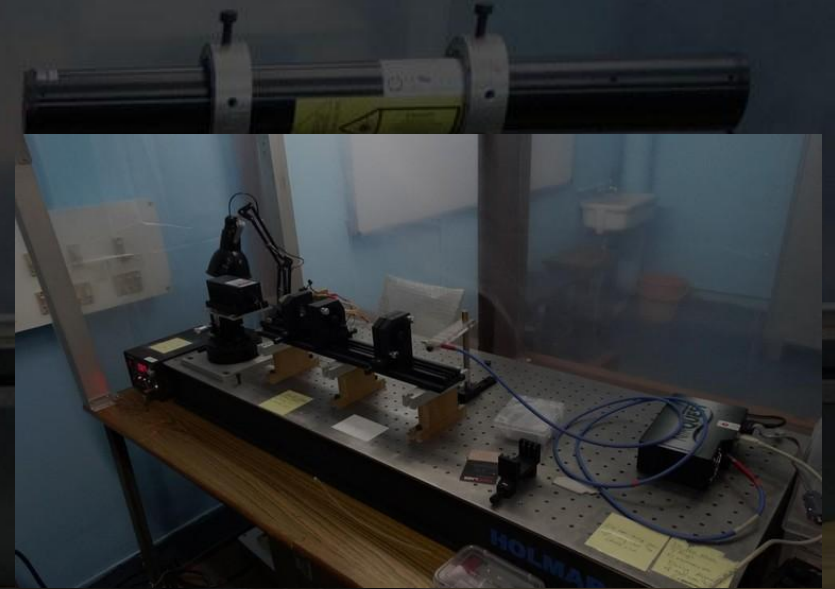
- fluorescence spectroscopy
- excited state photo physics and relaxation dynamics
- active-site protein dynamics

Microscale Transport Laboratory

- MICRO PIV/LIF
- Confocal Microscopy
- Digital Holography
- Laser/Color Schlieren
- PIV
- Interferometry
- Femtosecond Laser Microfabrication

Optical Instrumentation Lab

- Optical Techniques
- Flow control hierarchical porous medium



fs Laser Spectroscopy Lab



Microscale transport Lab

Our Alumni



Barada Kanta Nayak

M.Tech.(1999)
CEO at Spikewell



G Rajshekhar

M.Tech.(2006)
Professor, Department of Electrical
Engineering & CELP, IIT Kanpur



Pankaj Dhar

M.Tech.(1990)
Founder and CEO at A-Bioanalytics



Aveek Dutta

M.Tech.(2007)
Scientist, Government of India



Tripti Srivastava

M.Tech.(1994)
Chief TQM & CQA designate at Tata Steel



Dr. Prasad Chaphekar

M.Tech.(1994)
Deputy Secretary, Ministry of New and
Renewable Energy, Government of India

Our Alumni

Mukund Anandan

M.Tech.(2013)
Tech Lead (Software) at Cisco

Raju Ranjan

M.Tech.(2006)
Senior Consultant (AI) at
Datalens Technologies

Anup Lal Shah

M.Tech.(1990)
Scientist E, LASTEC, DRDO

Lakshay Thukral

M.Tech.(2021)
Sr. ASIC Engineer at NVIDIA

Tarique Anwar

M.Tech.(2021)
Staff R&D Engineer, Circuit
Design & Arch, Synopsys Inc

Nariseti Sundar Raju

M.Tech.(1999)
CTO at Spikewell

Aditya M.

M.Tech.(2021)
Multimedia Systems Engineer,
Qualcomm

Suchismita Bose

M.Tech.(2021)
ASIC Engineer at NVIDIA

Prakash Sahu

M.Tech.(2007)
Assistant Design Engineer at
Mecon Limited, India

Top Recruiters



and many more...

Contact Information



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